

Causal inference in latent space for time series and text data

IA pour la Science

16 february 2026

Introduction

The goal of this document is to describe the target function of the causal graph.

We treat the algorithm in the paper PCGCE in the paper "Extended summary graphs (ESG)" belongs to class of the exhaustive search algorithms. So to an optimal causal graph we make no difference between 1) exhaustive printing or combinatorial algorithms, 2) gradient-based algorithms including algorithms with relaxations, 3) zero-gradient algorithms. The only thing we expect its result is a causal graph, optimal according to this criterion.

The model structure includes: 1) the time is continuous $\frac{dx}{dt} = f(x, t)$ (basic: discrete with $\Delta t = 1$), 2) the time series X, Y, Z has a spacial structure $\frac{dx}{dx} = g(x, s)$ (basic: scalar-valued), 3) there is no limitations of the time as in the ESG, 4) the model is linear in the latent space (basic: linear in the observable space), 5) the latent space represent a spectrum of the observable domain.

The graph structure: 1) the graph is a bipartite graph between the observable and the latent, 2) the direct graph is isomorphic to the reverse graph.

The time structure for the convolutional model $\int_{\tau=0}^T f(\tau)h(t-\tau)d\tau$ 1) the time in the convolutional model is 2)

The space: